GS SECTOR SPECIALISTS: TMT

GS US TMT: .. 2026 TMT Outlook Note ....

Another solid year for TMT with the NDX up 20%+ in 2025, set to deliver back-to-back-to-back 20%+ annual returns for just the second time in 25+ years (.. or, for another factoid: ... the NDX only has 3 down years in the last 23 years..)

That said, it was not 'easy' sledding out there in 2025 as the first 4-months of the year featured several gut checks (**DeepSeek** in Jan, a **nasty Mo' unwind** in Feb/Mar and **Liberation Day** in April), **followed by, a low vol & alpha supportive move from May-Aug as most "themes" went right way during this window** (think **Semis up 12 of 13 weeks April-July**) **… however .. the going again got tougher down the stretch of the year** (felt like Sept-Dec was a "treading water" period for many) **as the market broadened out** (Fins, HC, Consumer & Industrials all traded well in 4Q) **and investors increasingly debated if NTM investments in GenAI would deliver the same, positive returns that LTM investments have thus far** (.. e.g. investors debating the path to a potential of $3 -4 tn in annual AI infrastructure spending by 2030, per Nvidia ..)

As we head into 2026, there is a healthy debate in markets if the recent "broadening out" dynamic is a durable sign of what's to come as the market embraces better cyclical visibility into the 1H'26 (see GS econ view below), **while also, searching for second (and third) derivative AI plays as we enter year 4 of the cycle [e.g. is there a better return in owning the "picks and shovels" at this point .... OR ... finding a name trading a discount that can leverage AI to cut cost or improve product discovery or drive new revenues streams?) ...**

... contra-side to this broadening out view would be an argument the Big Tech & AI Infra names have de-rated to levels where revisions (& compounded growth) can now do the heavy lifting (see MU px action re the power of revisions)**, model progress (and productization) is still very much ahead of us and these GenAI "sentiment swings" have become a feature – not a flaw – of this GenAI cycle..**

given all of that, my view on what lies ahead? ..... *strong / different views welcome*.

- I think the **return profile for the NDX should (*again*) be 'solid' next year – though, it wouldn't surprise me if returns were more 1H weighted** – due to an **arguably helpful starting place** (the NDX has been consolidating for a few months with a healthy "wall of worry to climb" re AI) **and given the potential unknowns that await into the 2H'26** (NFP data / jobs growth, mid-term elections, 2027+ AI visibility, pot'l weaker US economy) ..

- Once again, the **dominant theme will likely be GenA.I.** – though, one can debate if those that **produce AI "upside"** (think margins or revenue streams) will be rewarded more than those that **are "suppliers" of the AI** (semis, clouds, LLMs, etc).

- I also think there are number of **"fallen angels" across Tech / Growth Tech that could be worth a look again** (e.g. once loved TMT stocks that are well off the highs / out of favor): **V / MA, DASH, RBLX, ANET, AMZN, CRM, SNPS, SE, etc.**

*** see below for an extensive rundown of ideas, talking points, themes and charts ***

Happy Holidays & THANK YOU for the partnership, support & dialogue with the GS TMT team.

2025 …. STATS & STAND-OUTS

2025 Best Performers (as of 12/23): **SNDK** / **Kioxia** / **Nanya** up >500%, **SATS** +370%, **Optical** (**Innolight**, **LITE**) >350%, **IREN** / **BE** >300%, **WDC** +290%, **MU** +230%, **STX** +225%, **HOOD** +220%, **TTMI** / **CIEN** +185%, **WBD** +175%, **CRCL** +166%, **PLTR** +155%, **LRCX** +143%, **SPHR** +133%, **FIX** / **W** / **APP** / **FN** +120%

2025 Worst Performers (as of 12/23): **CCOI** -72%, **TTD** / **GLOB** -69%, **RPD** -62%, **WPP Group** -60%, **AI** -59%, **WIX** / **FIVN** / **CRTO** / **IT** -50%, **NSIT** / **DUOL** / **PAYO** / **HUBS** -45%, **STUB** -42%, **CHTR** / **POWI** / **FDS** / **TENB** -40%, **MNDY** / **FOUR** / **FVRR** / **GDDY** / **ZBRA** / **MANH** / **BILL** -35%, **BAH** / TEAM / **OS** / **S** / **GTLB** / **Meituan** down >30%

Mag 7+ performance in 2025 (as of 12/23) … **GOOG** +65%, **AVGO** +50%, **NVDA** +40%, **TSLA** +20%, **MSFT** +15%, **META** +13%, **AAPL** +8%, **AMZN** +5% … -- into 2026, I would say most investor inbounds on my end have been on **AMZN, META** & **MSFT** as of late.

Subsector "sentiment" rankings .. desk view on most to least loved subsectors in TMT …… **#1** (by a decent margin) **Semis** (& **Networking / comms infra**) .. **#2** (with a down arrow) **Internet** .. **#3** **Media / Entertainment** (think: FOX, WBD, SPHR, TKO types) … **#4** Hardware (AAPL, EMS, Connectivity) .. **#6 Software** .. **#5 Payments** .. **#7 Telecom / Cable** ..

Dispersion .. the big story of 2025 was dispersion – dispersion **between single stock vol** (high) vs **index vol (**low) …. The **dispersion within Tech stock performance** (NDX was up >20% this year, yet 30%+ of the NDX finished in the red) .. the **dispersion of subsectors** (Telecom, Payments & Apps SW lagged) …

Notable, large-cap stocks that underperformed the index (in the spirit of laggards) .. **Analog** (ON, TXN), **CDNS / SNPS, DDOG** (flat on year), **ANET, PCOR, Visa / MA, AMZN, LYV, CSGP, DIS, PANW, ABNB, NFLX**

Big Tech .. the **big 6** (nvda, aapl, goog, msft, amzn, meta) **added a collective ~$3.5 trn in market cap this year**, impressive, albeit a step **down from the ~$5.4+ trillion in market cap** added in 2024 and the **~$4.8+ trillion of market cap** added in 2023 …

Layoffs ... the blog layoffs.fyi (link) **tracked ~122k layoffs in the Tech industry in 2025** (.. this compares to ~150k tech layoffs in 2024 vs the ~265k layoffs in CY23)

Tech spending = solid next year .. Gartner forecasts **Worldwide IT Spending to Grow +9.8% in 2026 to $6+ trillion** (.. for context, this compares to Gartners' initial 2025 forecast of +9.3% y/y growth ....)

random thought #1: the "**ACN**" vs "**CAN**" autocorrect gets me every. single. time.

.. **random thought #2:** the number of permutations of "conference call" formats out there -- .. the PR only, the shareholder letter, the slide deck, the published prepared remarks ... where is the guide?! (in the PR? On the call? In the deck that gets released during the call?) .. is the call at 430pm? 445pm? 5pm? 530pm? ..

Traffic ... Global Internet traffic grew 19% in 2025 (up nicely from the +17.2% figure in 2024), per Cloudflare's year-end review (link)

Volumes .. an average of 17.5b shares traded across the U.S. equity market each day this year. For context, this number was 10.8b shares in 2020.

Global .. it surprised me to read that many (most?) **major equity markets outperformed the US in 2025** in local and in USD terms (**Korea** +70%, **Spain** +50%, **Europe** up in the 30s, etc)

Size ... the 7 largest stocks in the S&P 500 (NVDA, AAPL, MSFT, GOOGL, AMZN, AVGO, META) account for **36% of S&P 500 market cap** and **26% of earnings**, and **should contribute 46% of index EPS growth in 2026** [gir]

This is always worth the time .. Google's YouTube published its annual montage: "**2025: A Year in Search**": ... https://www.youtube.com/watch?v=Vv_sjpclsZ8 ... **volume up (↑) and enjoy** ...

STOCKS TO WATCH into 2026

Stocks that I like / am watching into next year .. **TSM, AMZN, IFX, AMAT, SNPS, ANET, V/MA, DASH, TTWO, Adyen, DIS, PCOR** ... [for context, my list from last year's 2025 outlook was: ... MSFT, TSM, AT&T, LRCX, Visa, SQ, PINS, GPN, TTWO, MDB, TER, IFX, TEAM, AMZN, OKTA]

Notable GIR / Goldman Research Sell ratings (as of 12/23/25) **...** **EBAY, ETSY, FDS, HRB, RHI, AKAM, VEEV, CHTR, SMCI, UPST, PYPL, ARM, MKSI, TXN ...**

2025 outlooks ... GIR published a number of 2025 Outlooks the last few weeks ... **Hardware + Internet + Semis + Autos / Auto Tech + Telecom** -- see GIR's 2026 Outlook splash page

Most debated and/or 'prove it' stocks into 2025 (.. generally view these as good trading / alpha opportunities..) – **Analog Semis, MRVL, NOW, DELL, PANW, CRWD, UBER, META, MSFT, AMD, ANET, OTAs** (ABNB, EXPE, BKNG)**, RBLX, DIS**

10 QUESTIONS for 2026

1. Where **does the AI debate lead us in 2026?** … **Physical AI** (Robots, AVs, Glasses) …**'productivity' winners** .. **regulation**? .. **ROIC**? ..

2. What can (**Application) Software companies do to get their multiples back**? .. what Software debate / overhang, if any, shows up in the next 12-24-months? (seats? agents? utilization? commoditization? LLM competition?)

3. **AAPL**: What is the **Apple** story into 2026? .. **defensive growth**? … **AI story**? … **Foldable phones**? .. Why has **Apple app store growth been decelerating** (see GIR's / sensor-tower work link)?

4. What are the broad-reaching **ramifications of a commodity super cycle** (DRAM, HDDs, NAND …. throw in Gold, Silver, Copper)? **Any other areas of 'tightness'? who can absorb price vs who cant? Supply? when is "peak"?**

5. **Should GenAI drive more "efficiency" –** which, could mean **headcount reductions** …. **will the market view this as a positive** (sign of productivity) **or as a negative** (pressure on the economy / NFPs)

6. What are the **best names to buy amongst the Internet companies where investors are debate margins** (and/or competition) into 2026? [e.g. **META, DUOL, DASH, RBLX, EBAY, SE, MELI**, etc]

7. **'The Cycles'** …..do we get a turn in any cyclical cycles in 2026 across **Housing**? **CRE** / **Real Estate**? **ISM data** (~3 yrs below 50)? **Analog**? **Autos**? ..

8. Can **HW / Semi-AI stocks <u>again</u> pace the tape higher in 2026** …. or will debates on GMs or spending visibility or competition weigh on sentiment?

9. How does the markets' view of **LLMs evolve in 2026?** … commodity? Multi-player market? AGI vs ASI? Opensource? China-based models? productization or raw intelligence? ..

10. What is *nobody* talking about today that we will all be talking about in 2026? …. **Agentic commerce? .. the return of SaaS stocks? … AI Productivity / use-cases? …**

(POTENTIAL) 2026 THEMES TO WATCH – *what else?*

Memory + commodity super cycle

Mid-term elections

Events (world cup, Olympics, 250th America birthday)

Box office is back (Toy Story 5, Moana Live Action, Avengers 5, Star Wars)
The year "Agents" (?)

Stimulus (the One, Big, Beautiful Bill Act - OBBBA)

Physical AI / AI at the "edge" (robots, AVs, devices, glasses)
"Voice" as an AI enabler
Content Creation (& consumption) in the era of AI
Content creation vs Content curation
Vibe Coding and Vibe-Content Creation
China AI models / Tech Stack (see bloomberg story: "China Prepares as Much as $70 Billion in Chip Sector Incentives")

Credit & Debt
LLM model updates
Consumer / Internet x GenAI (app integrations)
Agentic commerce
TikTok
Space
Quantum Computing
AI "fakes"
ASICs vs. GPUs
'De-Regulation'
Stablecoins & Tokenization
AI bottlenecks (power? Labor? Chips?)

GTA VI

China as an Export nation

What "***faster, cheaper, better***" software, code development means...

AI vs jobs/layoffs (... if AI 'works', does that mean fewer jobs? If so, how does market react?)

Security (in the era of AI and/or Quantum)

Prediction Markets
De-globalization
GLP-1s
campus networking (WiFi-7 upgrades + campus switching refresh)

DATES TO WATCH in EARLY JAN'26

[ongoing catalyst watch .... new LLMs from the Frontier labs ... NVDA / AMD chips into China .. Supreme Court re: USMCA tariff agreements ..]

New Fed Chair announcement expected (reports): early-Jan

GS Clean Tech & Energy Conference: Jan 5-6

CES Tech Show: Jan 5-9

- **NVDA**: Jensen (CEO) keynote on 1/5 @ 4pm ET
- **AMD**: Lisa (CEO) keynote on 1/5 @ 930pm ET
- **MRVL**: Matt (CEO) fireside chat on 1/6 @ 12pm ET

TSM monthly sales data: Jan 9

US CPI data: Jan 13

World Economic Forum: Jan 19-23

MSFT: ***Microsoft AI Tour*** in New York City: Jan 21

The 40th Annual AAAI Conference on Artificial Intelligence in Singapore: Jan 20-27

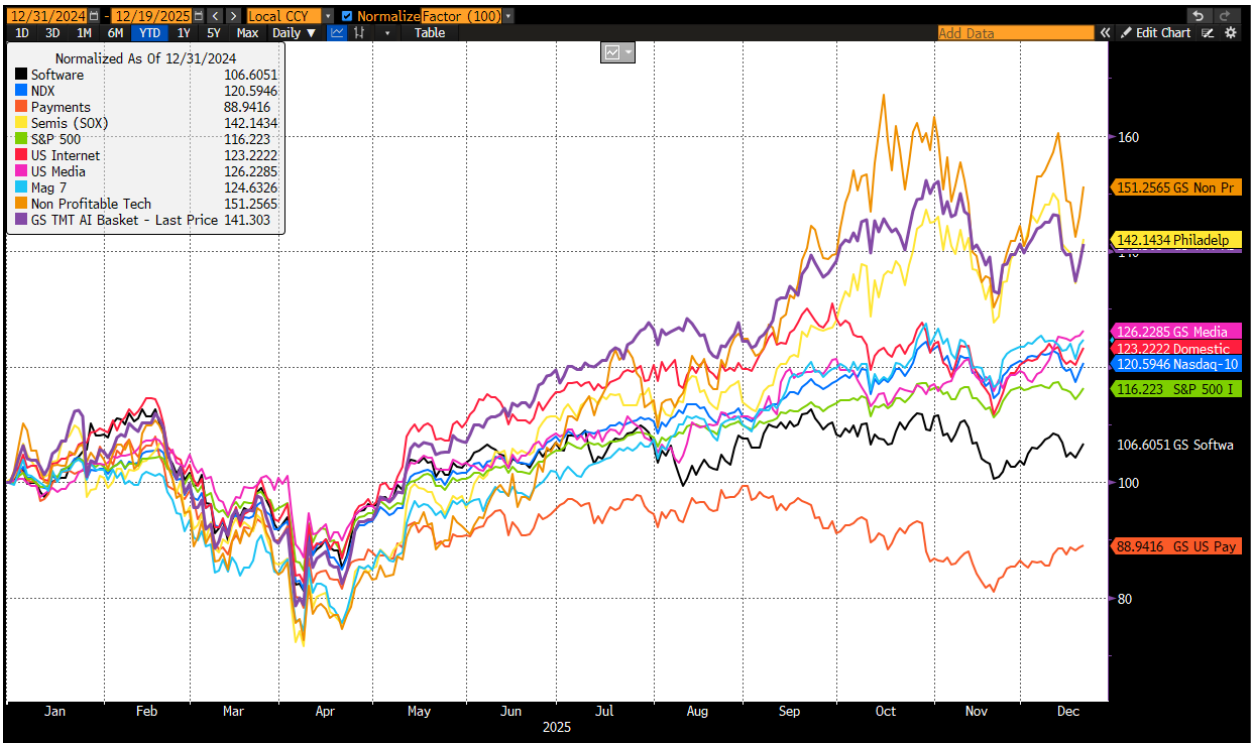
Megacap Tech 4Q earnings (MSFT, META, GOOG, AMZN): late-Jan

Jan FOMC meeting; Jan-28

Government shutdown deadline: Jan-30

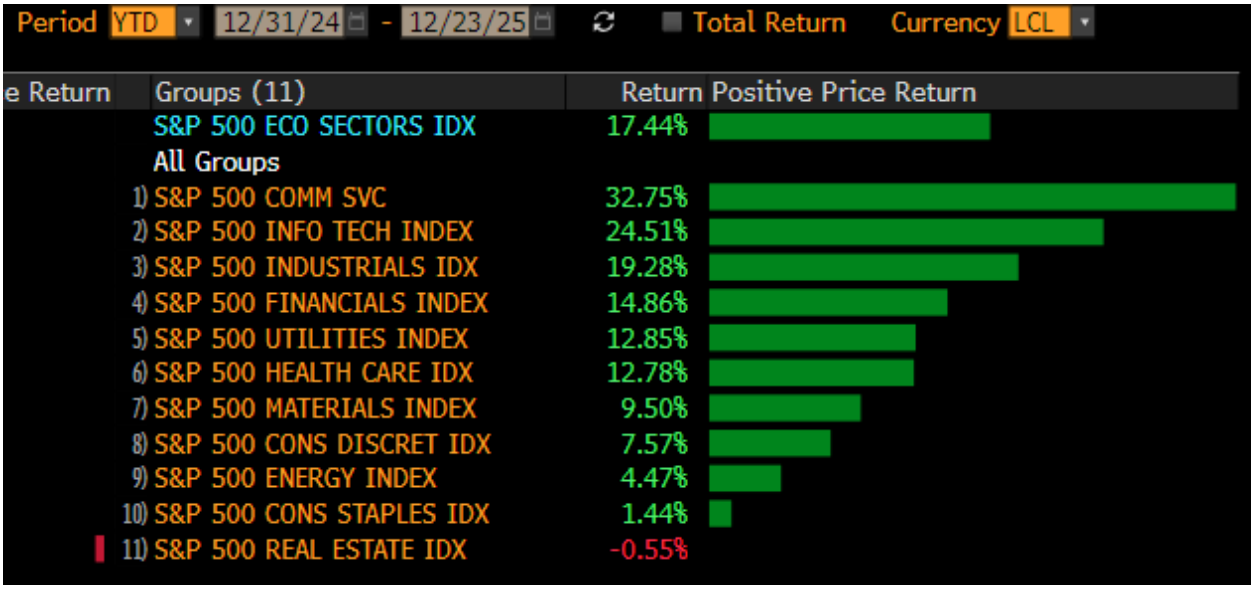
CHARTS in FOCUS

2025 end of year scoreboard in TMT ... Semis and AI stocks are set to finish the year up 40%+ .... vs .... Software +6% and Payments down ~10% ..



Source: bloomberg. GBM public. Past performance is not indicative of future results. 12/19/25

S&P500 … subsector performance .. while TMT Subsectors were out-in front in 2025, nearly every subsector finished in the black in 2025



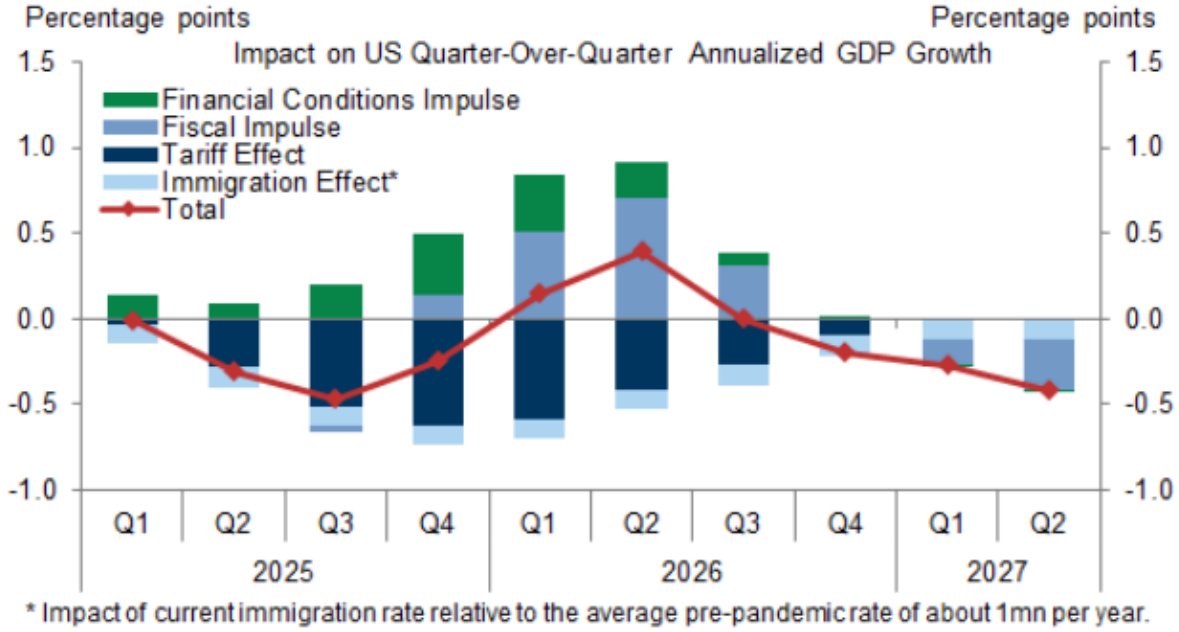
Period YTD 12/31/24 - 12/23/25 Total Return Currency LCL

e Return	Groups (11)	Return	Positive Price Return
	S&P 500 ECO SECTORS IDX	17.44%	
	All Groups		
1)	S&P 500 COMM SVC	32.75%	
2)	S&P 500 INFO TECH INDEX	24.51%	
3)	S&P 500 INDUSTRIALS IDX	19.28%	
4)	S&P 500 FINANCIALS INDEX	14.86%	
5)	S&P 500 UTILITIES INDEX	12.85%	
6)	S&P 500 HEALTH CARE IDX	12.78%	
7)	S&P 500 MATERIALS INDEX	9.50%	
8)	S&P 500 CONS DISCRET IDX	7.57%	
9)	S&P 500 ENERGY INDEX	4.47%	
10)	S&P 500 CONS STAPLES IDX	1.44%	
11)	S&P 500 REAL ESTATE IDX	-0.55%	

Source: bloomberg. GBM public. Past performance is not indicative of future results. 12/19/25

On the US Econ front, GS Econ / GIR house view is <u>upbeat</u> driven by a reduced tariff drag (as well as Gov't shutdown drag), **tax cuts from OBBBA, easing financial conditions … in turn, GIR expects especially strong GDP growth in the 1H of next year** [GIR ests real GDP growth of 3.5% in Q1 (qoq annualized) and 2.5% in Q2 before a sequential slowdown to 2.1% in Q3 and Q4]

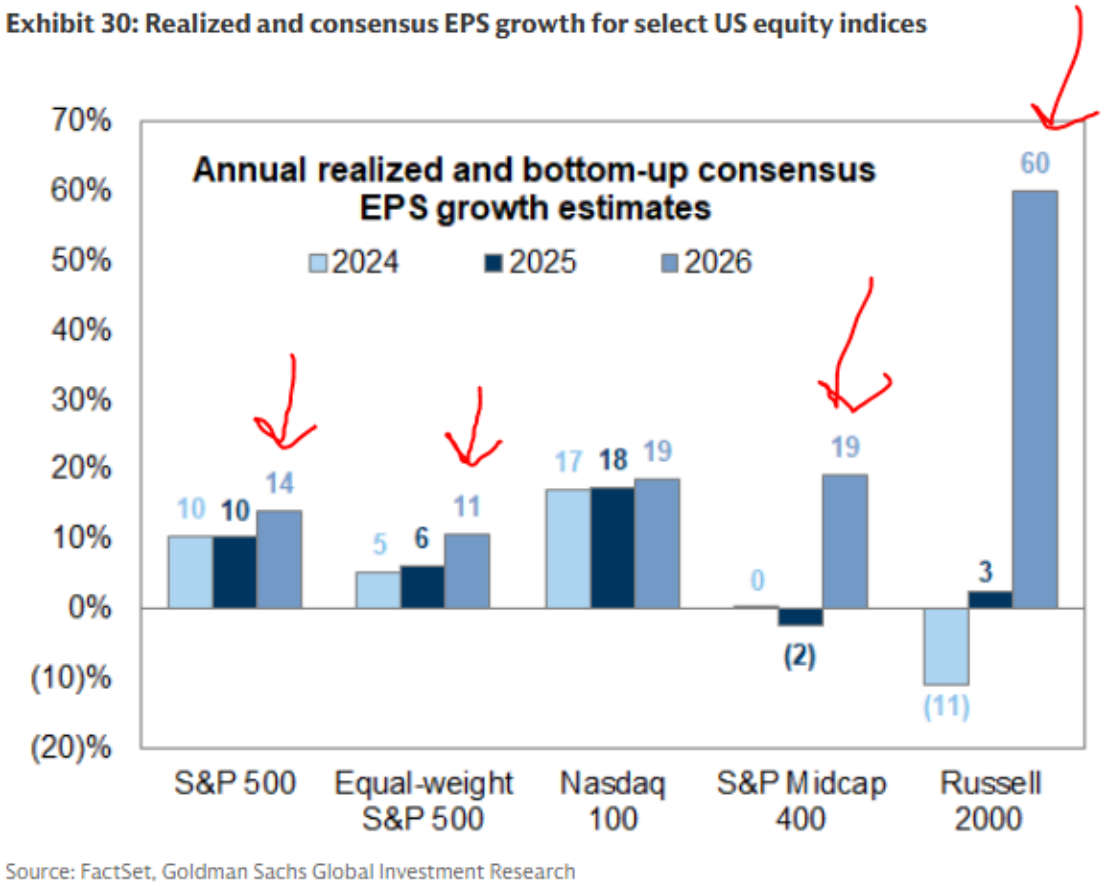
Exhibit 2: US Growth Likely to Benefit from Reduced Tariff Drag, Tax Cuts, and Easier FCI



Source: Goldman Sachs Global Investment Research

Source: https://marquee.gs.com/content/research/en/reports/2025/12/18/21ae60e7-6aa4-4e4c-a60a-0fa1a7d36300.html

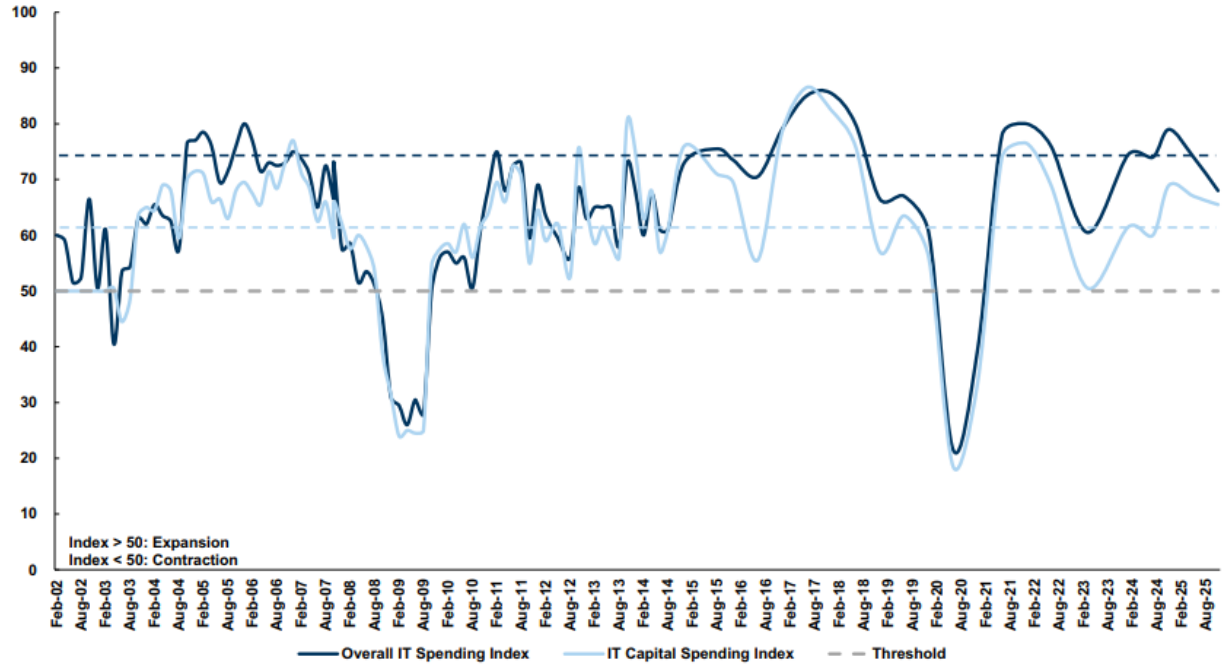
Is this likely to play out? … as I think about where the revisions could be in 2026 .. this chart from B Snider/GIR caught my eye …. <u>consensus expectations are calling for notable y/y accelerations in EPS growth everywhere outside of TMT (the NDX)</u> …. *is this likely to play out?* …. ***does MU's print serve as a reminder of the beat/raise potential in Tech?***



Source: GIR: https://marquee.gs.com/content/research/en/reports/2025/12/12/34ca9a9a-ebd8-45c4-99be-26251bdb8c26.html

2026 IT Spending Survey … IT spending intentions remain generally resilient in GIR's recent November 2025 survey -- despite uncertainty in the global macroeconomic backdrop

Exhibit 2: Overall IT Spending remained at historically elevated levels



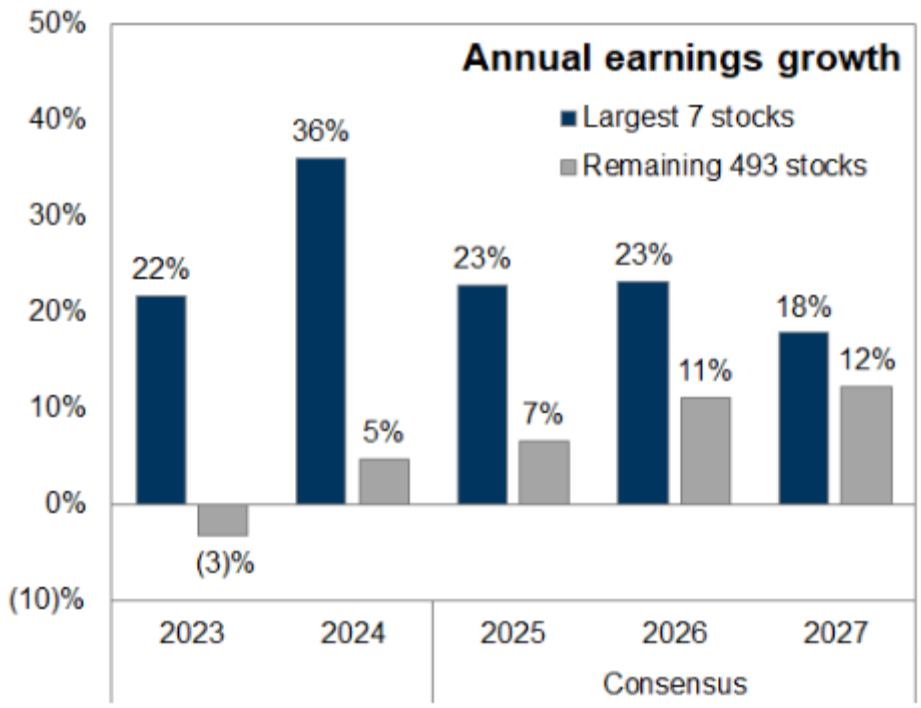
Index based on your current thinking, how do you expect your overall 2026 IT spending growth to be relative to 2025? What do you expect your IT capital spending growth (spending on new equipment and software only; not including staffing, services, depreciation, occupancy, or other) to be in 2026?

Source: Goldman Sachs Global Investment Research

Source: GIR. https://marquee.gs.com/content/themes/homepage-default.html

Mag 7 vs the field .. from Snider/GIR .. '*We expect earnings* ***growth for the seven largest stocks will account for 6 pp of the total 12% S&P 500 earnings growth in 2026****. In other words, firms accounting for 1% of S&P 500 constituents and 36% of market cap will account for 46% of index earnings growth next year. The* ***remaining 493 S&P 500 stocks will contribute 7 pp (54% of the total) to index level earnings growth****. We expect macro tailwinds from* ***accelerating economic growth*** *and a* ***fading tariff drag*** *on* ***margins will support an acceleration in the earnings growth rate for the remaining 493 stocks from +7% in 2025 to +9% in 2026.'***

Exhibit 9: Earnings growth for the largest 7 stocks and the S&P 493

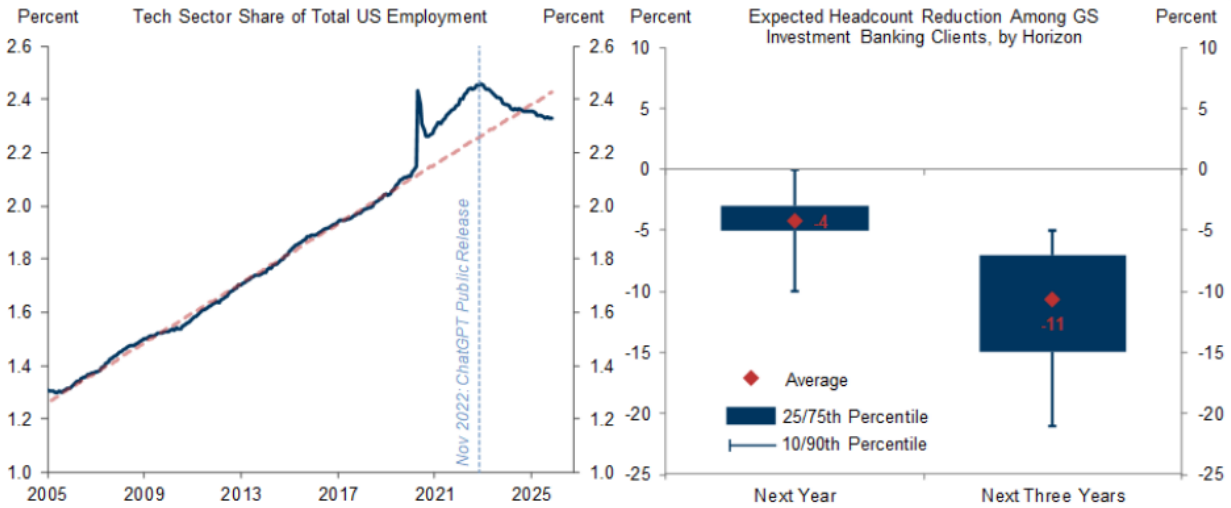


Source: FactSet, Goldman Sachs Global Investment Research

Source: GIR: https://marquee.gs.com/content/research/en/reports/2025/12/11/81a16af0-f242-4e57-a260-77ab8e77877a.html

How much of the productivity pickup and the associated weakness in the labor market is related to AI? .... The recent numbers don't change GIR's assessment that **the AI impact has so far been largely confined to the technology sector itself,** where the **left side of Exhibit 10 shows that employment trends have weakened materially in the last three years**

Exhibit 10: Impact of AI on Job Market Confined to Tech Sector So Far, But Broader Cost Cuts Loom



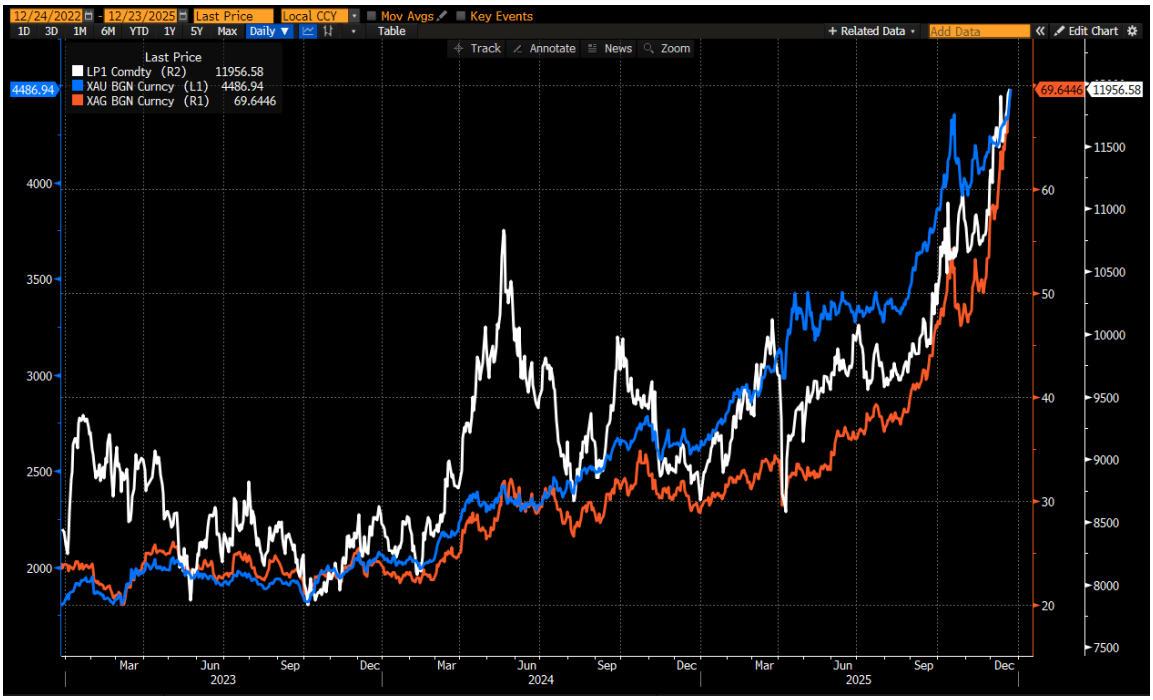
Note: Tech sector refers to the software publishers, data processing and related, web search and related, and computer systems design subsectors. Red line represents 2007-2019 trend.

Note: Results from survey of GS investment bankers, October 2025.

Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

Source: GIR: https://marquee.gs.com/content/research/en/reports/2025/12/18/21ae60e7-6aa4-4e4c-a60a-0fa1a7d36300.html

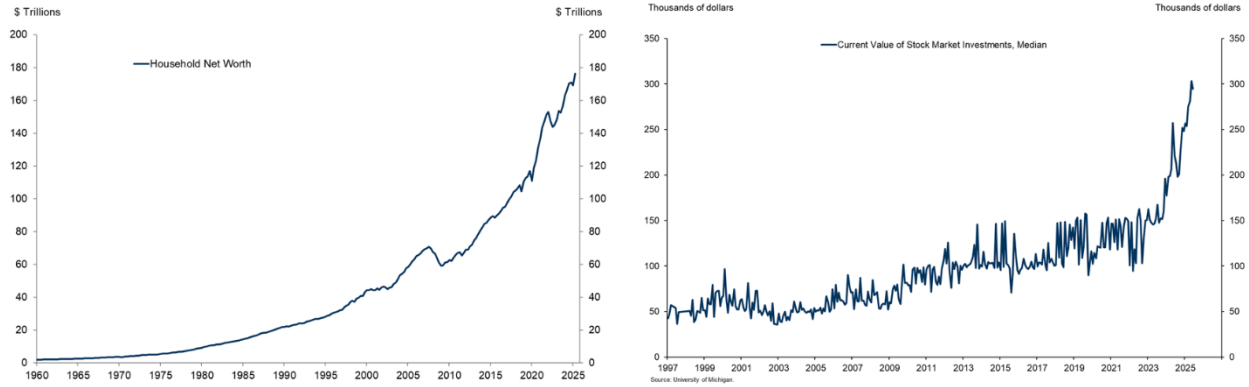
Commodity Super Cycle … while the Memory (HDD, NAND, DRAM – more below**) commodity super-cycle is well followed by Tech investors, it is also interesting to see Copper** (white), **Gold** (blue) and **Silver** (orange) **breaking out to all-time-highs..**



Rates.. something to keep an eye on into 2026 will be the divergence in Rates / Central Bank policy … as an example, Japanese 10yr Yields just hit ~20 year highs

Source: bloomberg. GBM public. Past performance is not indicative of future results. 12/19/25

Wealth .. here's the trend of US household net worth, expressed in nominal terms (through Q2) **…..** more recently … **the surge in the stock market has been a major contributor to the surge in household net worth** (source = Lexi Kanter via Tony P)



Source: GS econ. GBM public. Past performance is not indicative of future results. 12/19/25

Software vs the field … chart below of the Software (gs basket with ~75 names) **relative to the NDX … while we are at ~5 year low, the pair appears to be trying to bounce at a similar 'level' that it did in mid-2024**

Source: bloomberg. GBM public. Past performance is not indicative of future results. 12/19/25

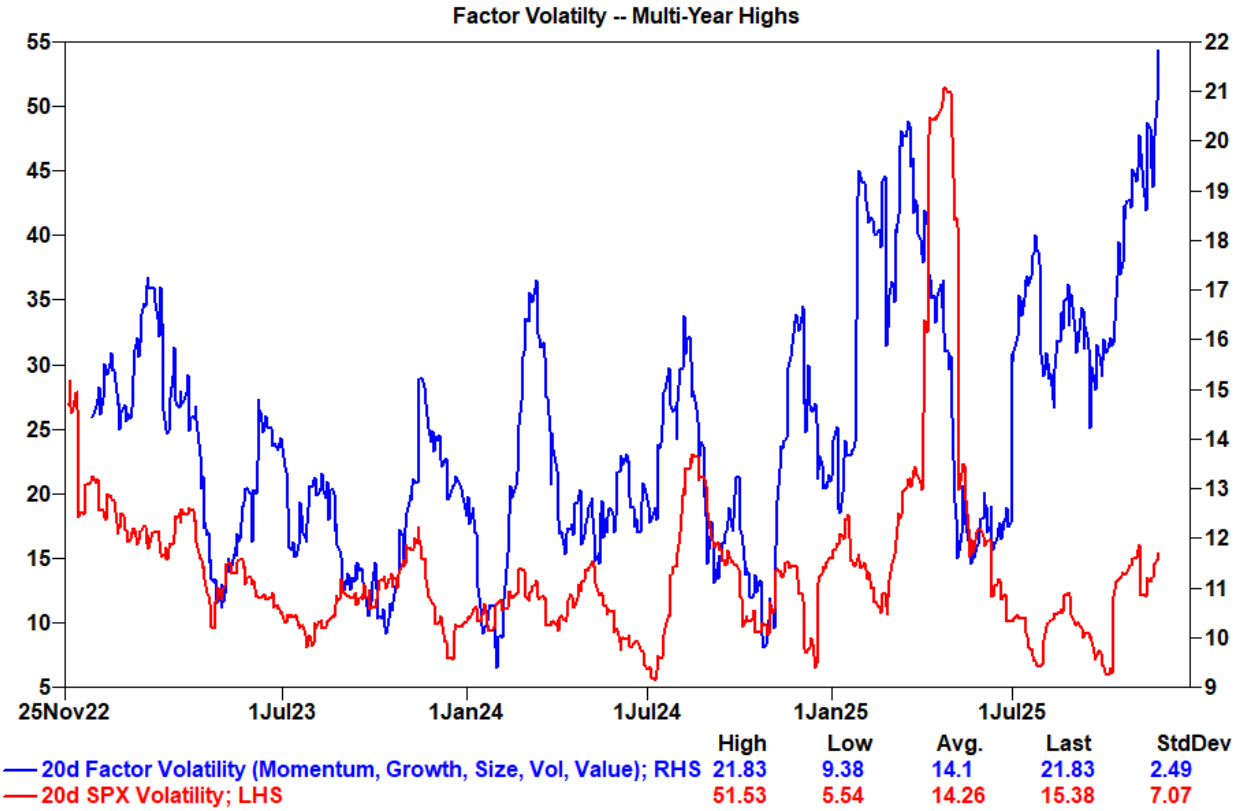
A life of leisure … handful of travel / leisure stocks are starting to break-out (**EXPE**, **CCL** and the **Hotel subsector GICs** – pictured below) … **a sign of Consumer optimism growing into 1Q ?? .. . on the back of this, investor convos around Payments have started to pick up into 2026** (think; V/MA, AFRM, Adyen types)

Source: bloomberg. GBM public. Past performance is not indicative of future results. 12/17/25

Valuation check … the NDX is set to finish the year around ~25x fwd P/E, largely inline with its average multiple over the last ~7 years (dotted line below) ..

Source: bloomberg. GBM public. Past performance is not indicative of future results. 12/19/25

If it has felt wobbly out there the last few months, this picture is indicative ... Factor Volatility hit multi-year highs in recent months ...



AMZN .. handful of questions in recent weeks as to *why* Amazon trades so heavily (... feels like the lack of clear thematic attach / torque to AI or to Consumer cyclicals?) .. **I'd also just flag the below: December = (historically) a tough month, January = (historically) best month**

AMZN US Equity | Spread Builder | View | Edit | Seasonality Chart

Last Price | Local CCY | Spread | <Type security> | Last Price

10 Years Ending 2025 | Percent Change | Net Change | High/Low/Avg

Calendar Year | Trailing 12M | 01-Jan - 31-Dec | Monthly | Line | Heat Map | Securities/Lines | Map Options

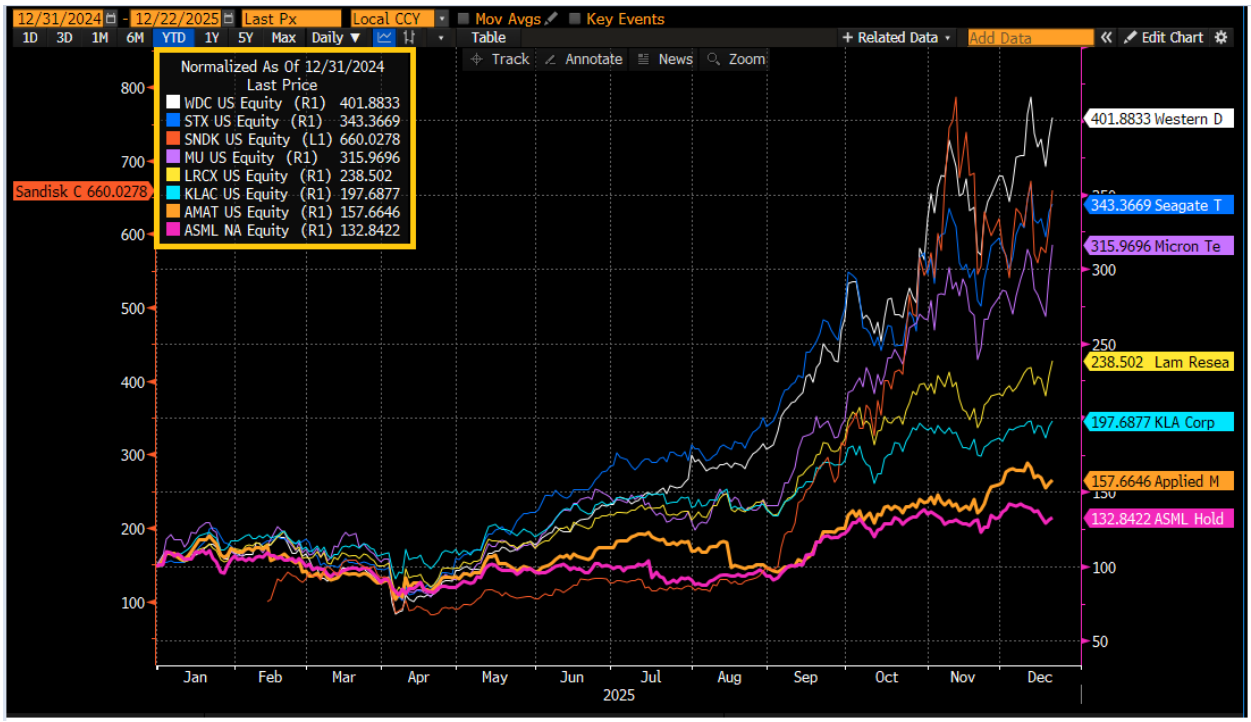
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
10 Yr Avg	7.12	.18	3.59	5.95	1.86	3.41	7.25	1.07	-2.45	.82	3.87	-1.31
2025	8.34	-10.69	-10.37	-3.07	11.16	7.01	6.71	-2.18	-4.12	11.23	-4.50	-4.56
2024	2.15	13.89	2.05	-2.98	.82	9.53	-3.24	-4.54	4.39	.04	11.53	5.53
2023	22.77	-8.63	9.61	2.09	14.35	8.11	2.55	3.24	-7.89	4.70	9.77	4.00
2022	-10.28	2.67	6.14	-23.75	-3.28	-11.65	27.06	-6.06	-10.86	-9.35	-5.76	-12.99
2021	-1.56	-3.53	.04	12.07	-7.05	6.74	-3.27	4.30	-5.35	2.66	3.99	-4.93
2020	8.71	-6.22	3.50	26.89	-1.28	12.96	14.71	9.05	-8.76	-3.58	4.34	2.81
2019	14.43	-4.59	8.59	8.19	-7.86	6.68	-1.42	-4.85	-2.27	2.35	1.36	2.61
2018	24.06	4.24	-4.30	8.21	4.05	4.31	4.57	13.24	-.48	-20.22	5.77	-11.13
2017	9.82	2.62	4.91	4.34	7.53	-2.68	2.04	-.73	-1.96	14.97	6.47	-.62
2016	-13.15	-5.87	7.44	11.11	9.58	-.99	6.04	1.36	8.86	-5.67	-4.97	-.09
2015	14.24	7.23	-2.12	13.35	1.77	1.13	23.51	-4.34	-.19	22.27	6.22	1.67

Source: bloomberg. GBM public. Past performance is not indicative of future results. 12/17/25

NVDA ... this is the normalized, annual performance of NVDA over the last few years (2025 = blue, 2024 = green, 2023 = yellow) ... **while obviously hard to make much of 3 years of price action, if you squint, you can arguably see a "pattern" of solid performance 1H performance vs more consolidation in the 2H ...**

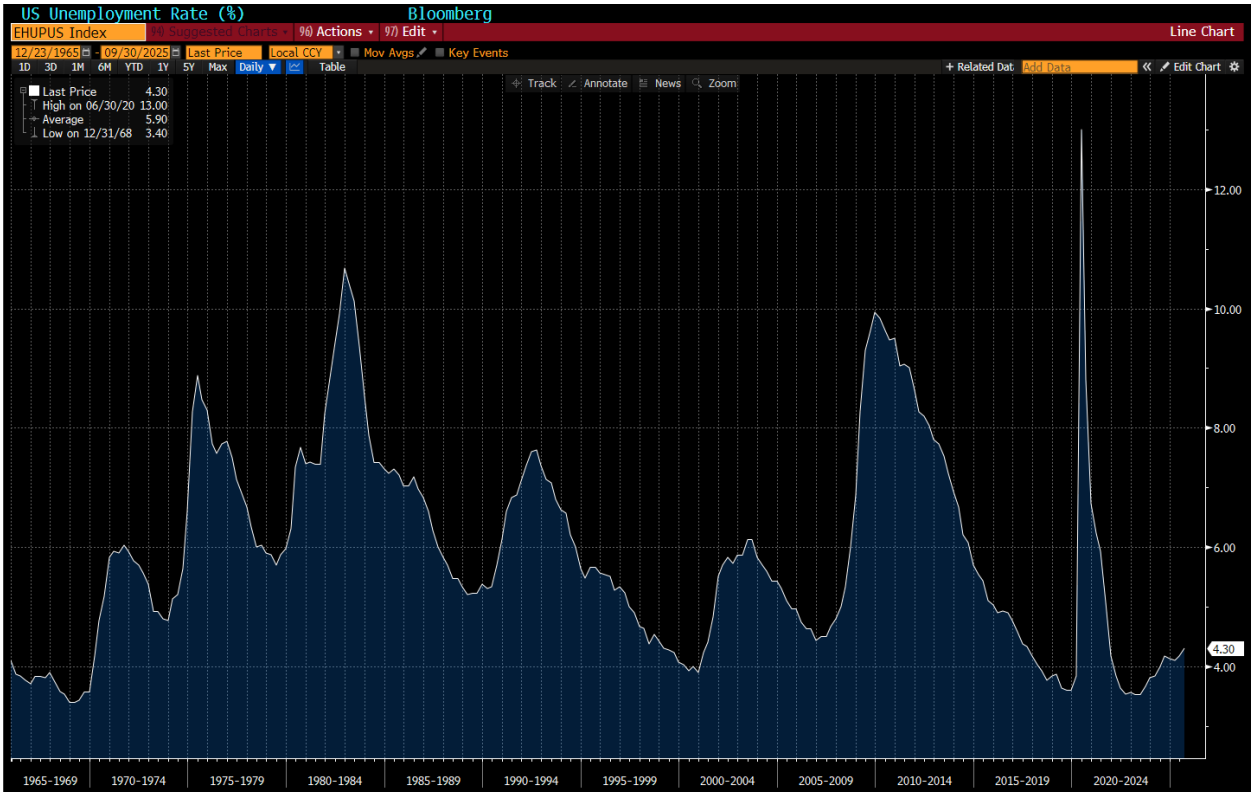
Source: bloomberg. GBM public. Past performance is not indicative of future results. 12/17/25

Unlike the others ... this year has been all about Memory (DRAM / NAND / HDD up ~3x-6x), and as a derivative, Semicaps (LRCX & KLAC are up >2x) .... what also jumps out to me are the laggards -- ASML +30% and AMAT +50% ... [yes, GIR is buy rated on both of them].



Source: bloomberg. GBM public. Past performance is not indicative of future results. 12/17/25

Jobs. A simple, yet important chart to keep an eye on into 2026+ is the US Unemployment rate -- think: the history of upward "trends" in the UE rate vs the rise of GenAI in corporate America



Source: bloomberg. GBM public. Past performance is not indicative of future results. 12/17/25

Peter Callahan

TMT Sector Specialist

Goldman Sachs FICC and Equities

212-902-2659